
Depth-Aware Selective Adaptation for Pretrained EEG Foundation Models

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Abstract

Pretrained EEG foundation models are beginning to show promising transfer behavior, but downstream adaptation is still dominated by either shallow classifier replacement or uniform dense finetuning. We study post-pretraining adaptation as a modeling problem and propose a depth-aware selective adaptation framework for pretrained EEG backbones. The method augments a pretrained CBraMod encoder with lightweight spatial and spectral-temporal specialist modules, routed using AttnRes-derived depth summaries and optional compact EEG context. Across five heterogeneous EEG benchmarks spanning affective, physiological-state, clinical-event, and motor-imagery transfer, we evaluate the method under matched downstream protocols rather than through new pretraining. The clearest practical gains appear on heterogeneous affective and physiological-state tasks, while seed-complete control comparisons show that AttnRes and specialist adaptation capture the most stable gains and depth-aware routing provides a smaller, more seed-sensitive refinement. A focused PhysioNet-MI analysis further identifies a boundary regime where simpler or more regularized adaptation can be competitive. These results support a cautious but useful conclusion: selective adaptation is promising for EEG foundation models, but its value is dataset-dependent and should be evaluated as part of the downstream transfer protocol.

1 Introduction

Electroencephalography (EEG) is a flexible and non-invasive neural recording modality with applications spanning affective computing, sleep staging, clinical event detection, neurodiagnostics, and brain-computer interfaces. Recent self-supervised and cross-dataset EEG pretraining results make reusable backbones increasingly plausible, but downstream adaptation remains underdeveloped: once a pretrained encoder is available, it is still unclear how to adapt it reliably across heterogeneous downstream tasks with different montages, sampling rates, label structures, and task semantics [Roy et al., 2019, Bommasani et al., 2021].

Current downstream practice is still dominated by a few simple extremes. One option is to freeze the pretrained backbone and train only a lightweight classifier head, which is efficient but may under-adapt under large task shift. Another is to densely finetune the whole backbone, which is expressive but expensive, structurally agnostic, and potentially unstable across heterogeneous EEG tasks. More broadly, parameter-efficient transfer methods and generic MoE-style adaptation provide useful recipes in other domains, but they do not explicitly exploit how pretrained EEG backbones organize information across depth, channels, and spectral-temporal structure [Houlsby et al., 2019, Li and Liang, 2021, Hu et al., 2022, Shazeer et al., 2017, Lepikhin et al., 2021, Fedus et al., 2022].

This paper proposes a depth-aware selective adaptation framework for pretrained EEG foundation models. Starting from a pretrained CBraMod backbone, we augment upper layers with typed spatial

and spectral-temporal specialists and route them using AttnRes-based depth summaries, with optional compact EEG context as an auxiliary cue. The key idea is to treat downstream adaptation as a structured modeling problem: adaptation capacity should be allocated selectively and should respect the representation geometry of the pretrained EEG backbone rather than being inserted as generic conditional computation.

The empirical goal is equally cautious. We evaluate the method under three evidence roles rather than collapsing all datasets into one benchmark claim: contextual practical-payoff evidence, matched reproduced control evidence, and focused boundary-case evidence. This framing is important because it lets us ask not only where selective adaptation helps, but also where its gains become smaller, more seed sensitive, or unnecessary.

Contributions:

- We introduce a **depth-aware selective adaptation framework** for pretrained EEG foundation models that replaces uniform dense adaptation with routed lightweight specialists.
- We propose **AttnRes-derived depth-summary routing**, in which depth-wise residual or attention behavior is summarized and used as an adaptation signal rather than relying only on current-layer features.
- We design **typed spatial/spectral-temporal specialist adaptation**, aligning the adaptation mechanism with EEG’s structured representation geometry.
- We provide a **controlled evidence-tiered evaluation** spanning contextual payoff evidence (FACED/ISRUC), matched reproduced control evidence (SEED-V/TUEV), and focused boundary-case evidence (PhysioNet-MI), showing that AttnRes and typed specialists provide the most stable gains while depth-aware routing/context is useful but smaller and more seed-sensitive.

2 Related Work

2.1 EEG Representation Learning and EEG Foundation Models

EEG transfer has historically been difficult because downstream settings differ sharply in subject population, montage, sampling rate, label granularity, and task semantics [Roy et al., 2019]. Earlier end-to-end EEG architectures therefore tended to be task-specific, with separate pipelines for decoding, brain-computer interfaces, and sleep staging [Schirrneister et al., 2017, Lawhern et al., 2018, Perslev et al., 2021]. More recently, self-supervised EEG representation learning has expanded through contrastive, multi-view, and masked-prediction objectives, shifting attention toward reusable pretrained backbones [Mohsenvand et al., 2020, Kumar et al., 2022, Kostas et al., 2021, Chen et al., 2024, Wang et al., 2024a].

This trend now includes both cross-dataset biosignal encoders and larger EEG-specific pretrained models. Cross-dataset work such as BENDR and BIOT aims to learn broadly reusable sequence representations across heterogeneous neural or physiological corpora [Kostas et al., 2021, Yang et al., 2023]. EEG-specific backbone work has pushed more directly toward reusable brain-signal encoders through masked modeling, autoregressive objectives, structural inductive bias, codebook-based tokenization, and scale-aware architectural design [Chen et al., 2024, Cui et al., 2024, Wang et al., 2024a, Yue et al., 2024, Wang et al., 2024b, 2025, Zhou et al., 2025, Karantonis et al., 2025, Maksudova et al., 2025, El Ouahidi et al., 2025].

2.2 Downstream Adaptation, PEFT, and Routing

Although pretrained EEG backbones are becoming more capable, downstream adaptation is still often treated as a protocol choice rather than as a modeling problem. Recent evaluation, review, and adaptation studies have begun to ask whether these backbones transfer reliably under downstream fine-tuning, including large-brainwave foundation-model evaluation, generic time-series foundation-model transfer to EEG, structured adaptation for pretrained EEG encoders, distribution-shift adaptation, and broader synthesis of emerging EEG foundation-model evidence [Lee et al., 2025, Turgut et al., 2025, Gnassounou et al., 2025, Ma et al., 2026, Lee et al., 2026, Liu et al., 2026].

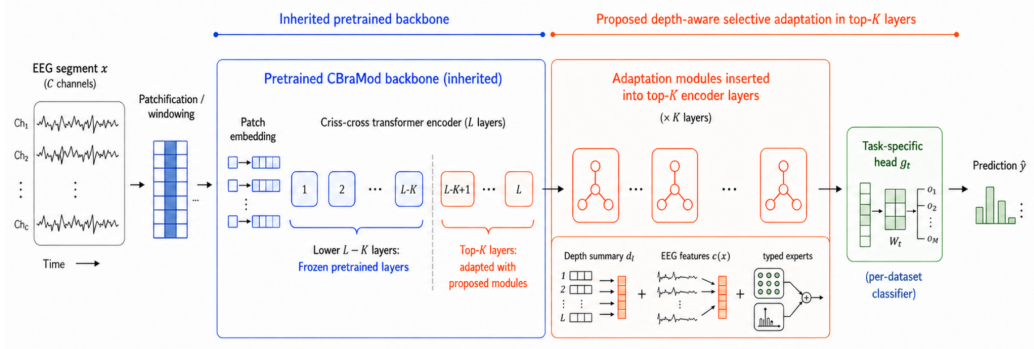


Figure 1: Overview of the proposed depth-aware selective adaptation framework. A pretrained CBraMod backbone encodes patchified EEG inputs, while upper-layer adaptation modules use AttnRes-derived depth summaries and, when enabled, compact EEG router features to route lightweight typed spatial and spectral-temporal specialists. The inherited backbone remains the primary representation engine; adaptation capacity is added selectively rather than through uniform dense finetuning.

In parallel, parameter-efficient transfer methods such as adapters, prefix-style tuning, and low-rank updates provide useful alternatives to dense finetuning in other domains [Houlsby et al., 2019, Li and Liang, 2021, Hu et al., 2022]. These methods motivate selective adaptation, but they do not by themselves specify which parts of a pretrained EEG backbone should receive additional capacity or which signals should control that allocation. Sparse routing and mixture-of-experts methods add a complementary idea: capacity can be allocated conditionally rather than updated uniformly [Shazeer et al., 2017, Lepikhin et al., 2021, Fedus et al., 2022]. This is attractive for EEG transfer, but generic PEFT and MoE recipes still do not specify how adaptation should respect backbone depth, channel structure, or spectral-temporal organization.

2.3 Why EEG Adaptation Needs Depth-Aware and Structure-Aware Routing

Backbones such as CBraMod and CSBrain improve EEG representation learning through stronger architectural inductive biases [Wang et al., 2025, Zhou et al., 2025], but they do not by themselves resolve the downstream adaptation problem. EEG representations are not formed uniformly across depth: spatial interactions, temporal context, and spectral cues are progressively integrated. This suggests that adaptation should not be governed solely by the current hidden state or applied uniformly to every layer. Our method builds on this observation by using AttnRes-derived depth summaries and typed spatial/spectral-temporal specialists so that routing reflects both backbone depth behavior and EEG-specific structure.

3 Method

3.1 Problem Setup

Building on the motivation in Sections 1 and 2, we now formalize the proposed adaptation mechanism. We study downstream adaptation of a pretrained EEG backbone. For an input trial $X \in \mathbb{R}^{C \times S \times T}$, where C denotes channels, S denotes patch or segment index, and T denotes samples within each patch, a pretrained backbone f_θ produces hidden representations $H^0, H^1, \dots, H^L$ with $H^\ell \in \mathbb{R}^{C \times S \times D}$. A task-specific head g_ϕ maps the final representation to a prediction $\hat{y} = g_\phi(f_\theta(X))$.

Our goal is not to redesign pretraining, but to improve downstream adaptation. Instead of uniformly finetuning the full backbone, we add lightweight selective adaptation modules to the upper transformer layers and route them using both current-layer content and depth-aware summary signals. This keeps the pretrained backbone as the main representation engine while allocating extra capacity only where it appears useful. Figure 1 presents the end-to-end pipeline.

3.2 Pretrained CBraMod Backbone

Our method reuses the pretrained CBraMod backbone as the base encoder. The key fact for this paper is that CBraMod builds representations on a channel–patch grid and separates channel-wise and patch-wise interactions within each criss-cross block. Because the inherited encoder already mixes temporal and coarse spectral structure, its hidden states are naturally organized along spatial, spectral-temporal, and depth dimensions. This is why our adaptation mechanism is structured around typed channel-wise versus patch/spectral-temporal specialists rather than around a generic unstructured adapter. Full patch-encoder details are deferred to the appendix.

3.3 AttnRes-Enhanced Depth Aggregation

To expose depth-wise information to the adaptation mechanism, we use AttnRes to aggregate hidden states from earlier depths. Let $\mathcal{S}_\ell = \{H^0, H^1, \dots, H^{\ell-1}\}$ denote the source pool available before layer ℓ . AttnRes produces a convex combination

$$\tilde{H}^\ell = \sum_{k=0}^{\ell-1} \alpha_k^\ell H^k, \quad \alpha_k^\ell = \text{softmax}_k(q^\top \text{Norm}(H^k)), \quad (1)$$

where the softmax is taken across depth. For notational simplicity, we show the operation as if each source produced one scalar weight, but in implementation the projection is applied over the hidden dimension and normalized across depth for each sample and token location. In practice, AttnRes can be inserted before attention, before the feed-forward transformation, or both; in our setting its most important role is to expose depth-selection behavior that can later guide routing.

3.4 Depth Summary and Router Context

Routing should not depend only on the current hidden state. We therefore derive a compact depth summary from AttnRes behavior. Depending on the variant, the summary is constructed from AttnRes selection weights, block-level projections, or compact statistics of depth-wise residual changes:

$$d^\ell = \Phi(\alpha_{\text{attn}}^\ell, \alpha_{\text{mlp}}^\ell) \in \mathbb{R}^r, \quad (2)$$

where Φ is a fixed summary operator and r is the router summary dimension. One summary variant is fixed per dataset for matched comparison; alternatives are reserved for ablations.

The router also receives a pooled content feature from the current layer state. When used, it may additionally receive a lightweight sample-level EEG descriptor $c(X) \in \mathbb{R}^m$, such as a pooled EEG summary or compact PSD summary. We keep this auxiliary term optional in the main formulation because compact EEG context is not part of every final reported recipe. The router input is therefore

$$z^\ell = \text{concat}(u^\ell, c(X), d^\ell), \quad (3)$$

with $c(X)$ omitted whenever compact context is disabled.

3.5 Typed Spatial and Spectral-Temporal Expert Routing

Selective adaptation is inserted only in the upper backbone layers. In each adapted layer, we replace the dense feed-forward update with a shared feed-forward path plus two specialist banks: a spatial bank and a spectral-temporal bank. The default implementation uses soft dispatch over each bank. We use “spatial” to refer to the channel-wise branch inherited from CBraMod, and “spectral-temporal” to refer to the branch associated with patch-wise interactions and optional spectral cues. Figure 2 zooms in on this selective adaptation block.

Let F_{shared}^ℓ denote the shared feed-forward mapping, and let $E_{\text{sp},j}^\ell$ and $E_{\text{st},j}^\ell$ denote the j th spatial and spectral-temporal experts. If π_{sp}^ℓ and π_{st}^ℓ are the router weights over the two banks, the adapted feed-forward output is

$$\mathcal{M}^\ell(H^{\ell-1}) = F_{\text{shared}}^\ell(H^{\ell-1}) + \gamma^\ell \left(\sum_j \pi_{\text{sp},j}^\ell E_{\text{sp},j}^\ell(H^{\ell-1}) + \sum_j \pi_{\text{st},j}^\ell E_{\text{st},j}^\ell(H^{\ell-1}) \right), \quad (4)$$

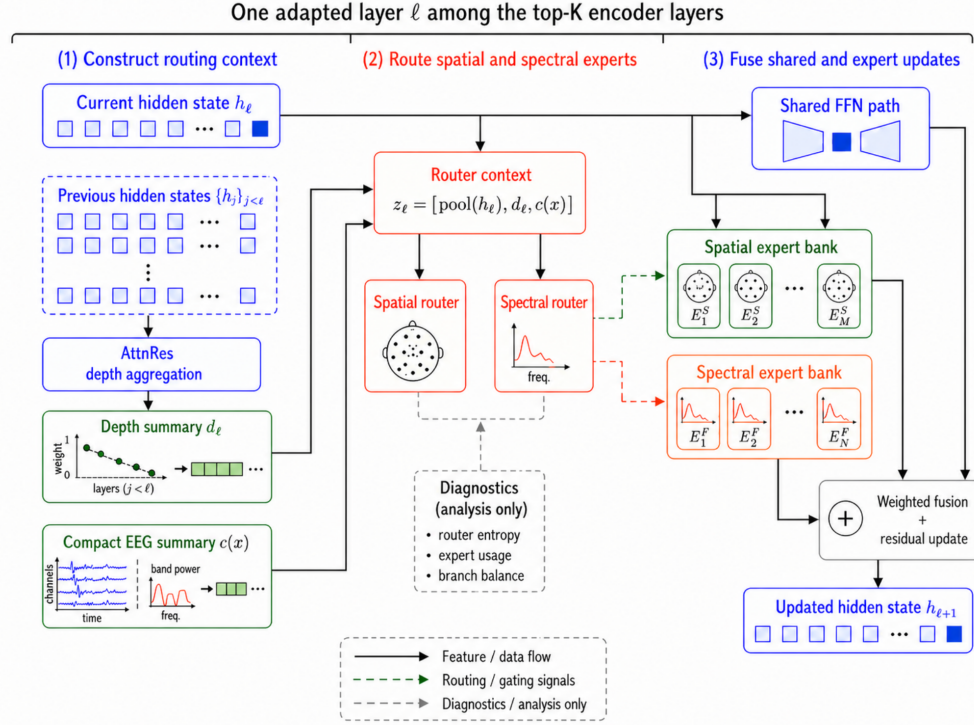


Figure 2: Selective adaptation block used in the upper layers. Routing depends on pooled current-layer content, optional compact EEG router features, and AttnRes-derived depth summaries. The router activates typed spatial and spectral-temporal expert banks, whose outputs are fused back into the shared backbone stream as residual adaptation paths.

where $\gamma^\ell \in [0, 1]$ is a residual scaling factor for the specialist paths. This decomposition is the key difference from generic MoE insertion: the model allocates adaptation capacity through two typed branches aligned with EEG’s spatial and spectral-temporal structure, and routing can depend on depth-aware summaries rather than only on local hidden content.

3.6 Training Objective and Diagnostics

The main training objective is the downstream supervised loss $\mathcal{L}_{\text{task}}$. When MoE adaptation is enabled, we add a small family of router regularizers for stability,

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}, \quad (5)$$

where the default $\mathcal{L}_{\text{aux}}$ uses entropy regularization, expert-balance regularization, and a small router-logit penalty. Additional routing variants, including hard-capacity dispatch, are treated as appendix ablations.

At validation time, the model exports routing diagnostics such as expert usage, routing entropy, effective number of experts, and branch-specific statistics. We use these diagnostics as supporting mechanism evidence while grounding the main claims in downstream performance.

4 Experimental Setup

4.1 Evaluation Suite

We evaluate downstream adaptation on five EEG benchmarks chosen to span distinct transfer regimes rather than to form a leaderboard suite. FACED [Chen et al., 2023] and SEED-V [Li et al., 2019] are multiclass emotion-recognition datasets, ISRUC [Khalighi et al., 2016] is a sleep-staging dataset,

Table 1: Evaluation suite and matched protocol summary. The table defines the downstream tasks, split policies, selection rules, and headline metrics used in the main paper. The key takeaway is that all five datasets are evaluated under prespecified per-dataset protocols rather than post hoc benchmark tuning. Full dataset statistics and preprocessing details are deferred to Appendix A.

Dataset	Task	Classes	Split protocol	Selection rule	Main metrics	Role in evaluation
FACED	Emotion recognition	9	Predefined train/val/test	κ	BA, wF1	Affective transfer
SEED-V	Emotion recognition	5	CBraMod-compatible split	κ	BA, wF1	Control / ablation anchor
ISRUC	Sleep staging	5	Subject-level 80/10/10	κ	BA, wF1	Physiological-state transfer
TUEV	Event classification	6	Fixed processed train/eval/test	κ	BA, wF1	Clinical event transfer
PhysioNet-MI	Motor imagery	4	Predefined train/val/test	κ	BA, wF1	Boundary-case transfer

TUEV [Obeid and Picone, 2016] is a clinical EEG event-classification dataset, and PhysioNet-MI [Schalk et al., 2009] is a motor-imagery classification dataset. This mix tests whether one pretrained EEG backbone can be adapted across affective, physiological-state, clinical-event, and motor-imagery settings.

4.2 Fairness Constraints and Matched Comparisons

All reproduced comparisons are run through the same downstream finetuning pipeline and are matched in preprocessing, pretrained CBraMod initialization, training budget, and checkpoint-selection rule within each dataset. We do not merge numbers obtained under different split policies or preprocessing conventions into one causal benchmark table. When previously reported values from external papers are used for context, they are clearly separated from reproduced matched comparisons and never used as the primary evidence for the adaptation claim.

The main paper therefore emphasizes confirmatory reproduced evidence. Multi-seed matched comparisons support the central benchmark table, while single-run or incomplete-seed ablations are reported as supportive or exploratory evidence in the appendix.

4.3 Baselines and Metrics

We compare the proposed method against adaptation baselines built on the same pretrained CBraMod initialization: dense finetuning, AttnRes-only adaptation, MoE-style adaptation without depth-aware routing, and the full depth-aware selective adaptation model. Frozen-backbone, top- k finetuning, and generic adapter-style controls are documented as baseline definitions and included in the main paper only when reproduced under matched protocol.

We report balanced accuracy (BA), Cohen’s κ , and weighted F_1 on all multiclass tasks. The main paper uses BA and weighted F_1 as the headline metrics, while the appendix reports the full BA/ κ /wF1 triplet together with per-seed values. Checkpoint selection is fixed within each dataset family through the corresponding run-level selection metric, which is made explicit in Table 1.

4.4 Optimization and Reporting Scope

Unless otherwise noted, downstream models are trained with AdamW, fixed epoch budgets within each matched family, and a prespecified validation-based checkpoint-selection rule. The codebase supports optional exponential moving average evaluation, component-wise learning-rate multipliers, and router regularization, but these are reported as named experimental choices rather than hidden tuning knobs. Appendix F records the exact dataset-specific settings and resource details used in the final reported families.

Table 2: Evidence-tiered headline performance across payoff and control regimes. Block A shows contextual payoff comparisons against reported CBraMod values. Block B shows reproduced headline comparisons against dense finetuning and AttnRes controls under matched downstream protocols, with reported CBraMod values included only as orientation context. Cells report BA / wF1 (%), and deltas are measured in percentage points (pp). Seed-complete mean $\pm$ std and per-seed values are reported in Appendix C.

Block A: Contextual payoff evidence					
Dataset	Regime	Evidence tier	Reported CBraMod	Ours full	Δ vs reported
FACED	affective transfer	contextual payoff	55.09 / 56.18	60.55 / 60.72	+5.46 / +4.54
ISRUC	physiological-state transfer	contextual payoff	78.65 / 80.11	80.68 / 83.06	+2.03 / +2.95
Block B: Reproduced headline evidence with reported context					
Dataset	Reported CBraMod	Dense	AttnRes	Full selective	Δ Full–Dense
SEED-V	40.91 / 41.01	40.24 / 41.11	40.81 / 41.64	41.62 / 42.57	+1.38 / +1.46
TUEV	66.71 / 83.42	62.13 / 81.88	61.15 / 80.14	67.98 / 81.89	+5.85 / +0.01

5 Results and Analysis

5.1 Strong Practical Gains on High-Impact Transfer Regimes

We begin with the downstream regimes in which selective adaptation is most visibly useful, while keeping the causal strength of each comparison explicit. Table 2 separates two evidence tiers rather than collapsing all datasets into one leaderboard. FACED and ISRUC provide contextual payoff evidence: the final selective recipe improves over reported CBraMod values in heterogeneous affective and physiological-state regimes, but these rows are not treated as causal proof because they are not part of one fully reproduced matched comparison matrix. SEED-V and TUEV provide the stricter reproduced evidence, comparing dense finetuning, AttnRes, and the full selective model under matched downstream protocols. Under this stricter view, SEED-V improves over dense in both BA and wF1, while TUEV shows a strong BA gain with mixed or competitive wF1. Appendix C reports seed-complete mean $\pm$ std and per-seed results for stability.

5.2 Control-Style Comparisons on SEED-V and TUEV

SEED-V and TUEV serve as matched reproduced control datasets. They ask the stricter question of whether selective adaptation improves over dense finetuning and simpler adaptation variants when the pretrained backbone, split policy, training budget, and selection rule are fixed. SEED-V remains the cleanest component-validation anchor: its gain over dense is modest, and much of the improvement is recovered before the final routing refinement. TUEV provides a second control-style family, where the full selective model achieves the strongest reproduced BA while wF1 remains mixed rather than uniformly better.

5.3 Component Ablation

Table 3 reorganizes the SEED-V ablation as a headline component-validation ladder rather than as a second benchmark table. We use the same headline-value convention as Table 2 and Figure 3 so that the main-paper evidence is internally consistent. The purpose is not merely to identify the strongest row, but to specify which modeling question each comparison answers. AttnRes accounts for the first improvement over dense finetuning, typed specialist capacity adds a smaller gain, and the full routing/context design provides the strongest headline refinement. Appendix C then gives the seed-complete mean $\pm$ std view, which shows that the earlier AttnRes and specialist-capacity gains are more stable than the final routing/context step. Thus, the appropriate interpretation is cumulative and cautious: the full method aggregates several structured adaptation choices, while the final routing refinement is useful but more seed-sensitive.

Table 3: Headline component ladder on SEED-V. The table reports the same headline reproduced values used in Table 2 and Figure 3, organized as a component ladder rather than as a second benchmark table. Deltas are measured in percentage points relative to the comparison named in the second column. Appendix C reports seed-complete mean $\pm$ std and per-seed values for stability; that view indicates that AttnRes and specialist capacity provide the most stable gains, while the final routing/context refinement is more seed-sensitive.

Question	Comparison	Component	BA	wF1	Δ (pp)	Stability
Depth aggregation	AttnRes vs Dense	AttnRes summary	40.81	41.64	+0.57 / +0.53	stable modest gain
Typed specialists	MoE-only vs AttnRes	spatial / spectral-temporal experts	40.95	41.74	+0.14 / +0.10	small gain
Depth-aware routing	Full-block vs MoE-only	depth-aware router	41.13	41.78	+0.18 / +0.04	seed-sensitive
Router context	Full-dual vs Full-block	dual-query context	41.62	42.57	+0.49 / +0.79	headline gain; seed-sensitive

Figure 3 visualizes the same headline component ladder and places it next to the TUEV matched subset and representative expert-utilization diagnostics. This keeps the visual story aligned with the main headline table, while the appendix provides the more conservative seed-complete stability view.

5.4 Boundary-Case Analysis on PhysioNet-MI

PhysioNet-MI tests a boundary regime rather than a headline gain. Figure 4 visualizes these focused ablations as zero-centered deltas, making the positive and negative effects relative to each reference condition explicit. Bank regularization plus component-wise learning-rate adjustment improves BA/wF1 relative to the regularization-free reference, whereas component-LR alone hurts and compact EEG/PSD router context remains below the no-context reference. Because these runs are not seed-complete matched families, we treat them as supportive boundary evidence: motor-imagery transfer may favor simpler or more regularized adaptation, even when richer selective routing helps in affective or physiological-state regimes.

5.5 Evidence Scope, Robustness, and Routing Diagnostics

Routing diagnostics are supporting mechanism evidence rather than standalone proof. SEED-V/TUEV logs show nontrivial branch-dependent expert utilization, suggesting that the specialist path is not trivially unused; fuller entropy, N_{eff} , and top-share diagnostics are in Appendix E. Overall, the evidence remains intentionally tiered: FACED/ISRUC support practical payoff, SEED-V/TUEV support matched-control interpretation, and PhysioNet-MI anchors the boundary case.

6 Limitations and Conclusion

6.1 Limitations and Future Work

This work is a controlled adaptation study rather than a new EEG foundation-model paper. The current evidence is CBraMod-specific and combines contextual payoff, matched-control, and boundary-case regimes rather than one uniform benchmark family. Broader seed-complete comparisons across datasets and backbones will be needed to test generality, especially for mechanism-level claims about the smaller, seed-sensitive routing/context refinement.

6.2 Conclusion

We introduced depth-aware selective adaptation for pretrained EEG foundation models, routing typed spatial/spectral-temporal specialists with AttnRes-derived depth summaries and optional compact EEG context. Across heterogeneous transfer regimes, the evidence supports a cautious claim: selective adaptation is useful but dataset-dependent, with the most stable gains from AttnRes and specialist capacity and smaller refinements from depth-aware routing/context. More broadly, this shifts EEG foundation-model evaluation from backbone-only comparison toward downstream adaptation design.

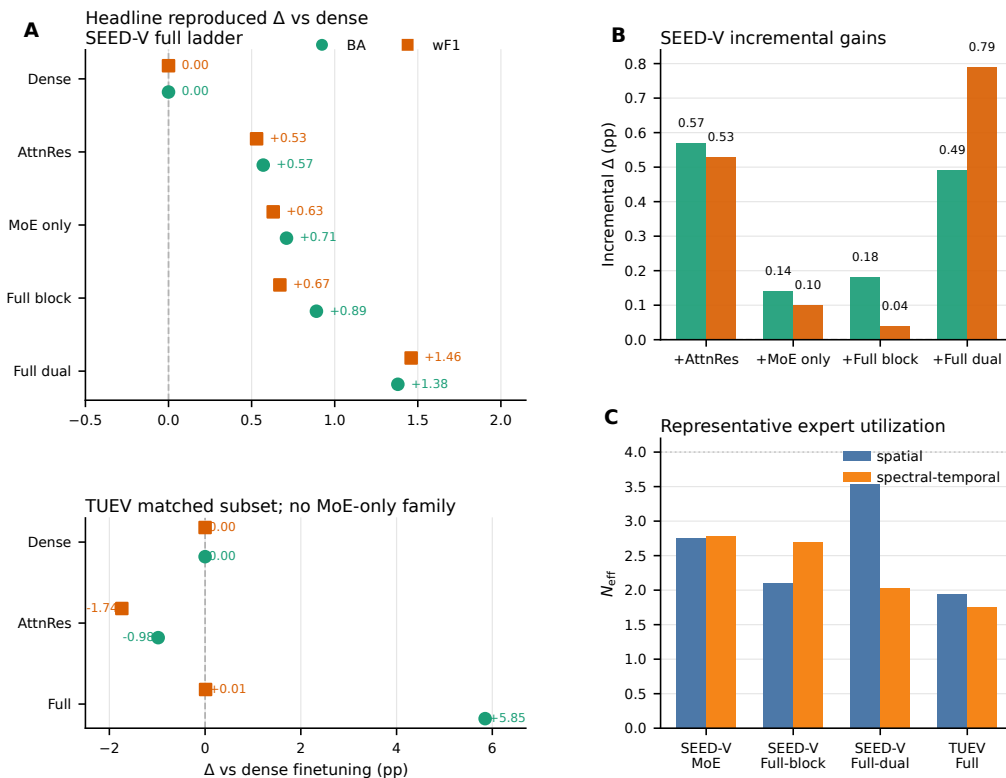


Figure 3: Headline component contribution and representative routing behavior. **A**: headline reproduced deltas relative to dense finetuning for SEED-V and TUEV; the TUEV subset reports the available matched family because a matched MoE-only run family is not available. **B**: SEED-V incremental gains show that the full model combines contributions from AttnRes, typed specialists, and routing/context design. **C**: representative expert-utilization diagnostics show branch-dependent routing behavior. Seed-complete mean $\pm$ std values and full routing-entropy diagnostics are reported in Appendices C and E.

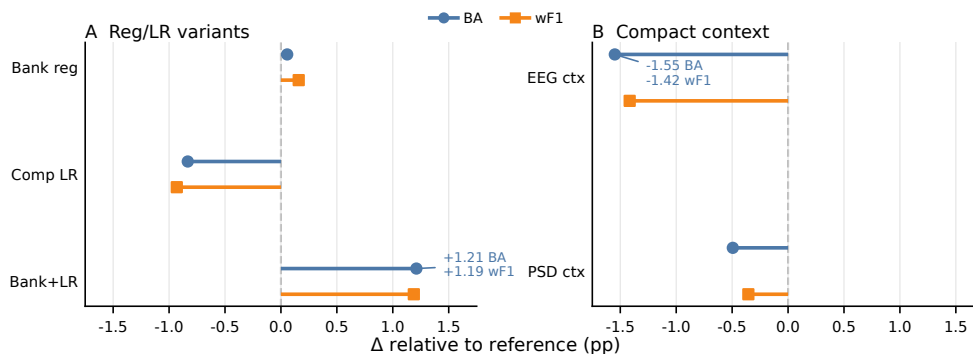


Figure 4: PhysioNet-MI boundary-case ablations shown as single-run deltas relative to the relevant reference. Positive values indicate improvement and negative values indicate degradation. Bank regularization plus component-wise LR helps in this focused setting, whereas compact EEG/PSD context does not improve over no-context routing. Full values are in Appendix D.

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Appendix Roadmap

This appendix is organized as a support document for the paper’s evidence trail. Appendix A establishes the protocol-clean evaluation suite and leakage guards. Appendix B maps the paper’s method names to implementation variants and final dataset recipes. Appendix C reports the seed-complete results behind the headline main-paper values. Appendix D collects extended ablations and boundary-case evidence. Appendix E reports routing diagnostics as mechanism-supporting evidence. Appendix F records reproducibility details and result-provenance summaries.

Table 4: Appendix roadmap. Each section is organized around a concrete evidentiary role rather than as a results dump.

Appendix aim	Appendix section	Evidence provided
Are the protocols clean?	Appendix A	Dataset scope, splits, preprocessing, leakage guards
What exactly was implemented?	Appendix B	Final recipes, variant definitions, trainable modules
Do headline values survive seed checks?	Appendix C	Best/headline values, mean±std, per-seed results
Which ablations shape interpretation?	Appendix D	Context variants, payoff variants, PhysioNet-MI boundary cases
Is routing actually used?	Appendix E	N_{eff} , $H/\log E$, top-expert share, branch diagnostics
How can the reported runs be audited?	Appendix F	Hyperparameters, compute envelope, provenance summaries

A Dataset Protocols and Leakage Controls

Purpose. Document the protocol-clean evaluation suite, preprocessing, split policy, and leakage guards so that the main-paper comparisons can be interpreted consistently within each retained dataset family.

A.1 Protocol-clean evaluation suite

The protocol-clean suite contains FACED, SEED-V, ISRUC, TUEV, and PhysioNet-MI. The retained datasets are intentionally heterogeneous and serve different interpretive roles: FACED/ISRUC support practical payoff, SEED-V/TUEV support matched control interpretation, and PhysioNet-MI anchors the boundary-case analysis.

Table 5: Dataset scale, task structure, and signal format for the protocol-clean suite. The retained datasets span distinct downstream tasks, recording formats, and evaluation roles.

Dataset	Task	Subjects	Samples	Classes	Rate	Channels / Segment	Role
FACED	Emotion recognition	123	10,332	9	250 Hz	32 / 10 s	contextual payoff
SEED-V	Emotion recognition	16	117,744	5	1000 Hz	62 / 1 s	matched control
ISRUC	Sleep staging	100	89,240	5	200 Hz	6 / 30 s	contextual payoff
TUEV	Event classification	370	112,491	6	256 Hz	16 / 5 s	matched control
PhysioNet-MI	Motor imagery	109	9,837	4	160 Hz	64 / 4 s	boundary case

A.2 Dataset scale, signal format, and preprocessing

These tables document the heterogeneity of the retained protocol-clean suite. Dataset scale and label structure differ substantially across affective, physiological-state, clinical-event, and motor-imagery

settings, and signal formats vary in sampling rate, channel count, and window duration. Within each dataset family, however, all compared methods share the same serialized data, segmentation, and normalization policy.

All datasets use fixed serialized preprocessing within each family: FACED uses repository processed 10 s segments, SEED-V uses the LMDB export from the SEED-V branch, ISRUC uses the processed filter-35 sleep pipeline with 30 s epochs, TUEV uses the standard processed benchmark release with 5 s event windows, and PhysioNet-MI uses repository processed 4 s trial windows. Loader-level channel scaling is used throughout.

A.3 Split policy, leakage guards, and selection

The split and selection policies below define the evaluation contract for each dataset. Selection is prespecified through a validation metric, and leakage guards are dataset-specific. These details are why the main paper separates within-dataset matched comparisons from contextual reported values rather than combining everything into one causal leaderboard.

Table 6: Split and selection policy by dataset. Each dataset uses a prespecified split and validation-based selection rule, preventing headline values from depending on ad hoc checkpoint choice.

Dataset	Split policy	Leakage guard	Selection rule
FACED	predefined train/val/test split	subject-disjoint predefined split	best validation κ
SEED-V	predefined CBraMod-compatible split	fixed split reused across methods	best validation κ
ISRUC	subject-level 80/10/10	subject-wise separation	best validation κ
TUEV	fixed processed train/eval/test folders	canonical processed benchmark split	best validation κ
PhysioNet-MI	predefined train/val/test split	fixed processed subject-level split	best validation κ

These protocol details explain why the paper does not collapse all datasets into one undifferentiated benchmark family. The within-dataset comparisons are clean, but the retained families serve different argumentative roles in the paper and appendix.

B Implementation Details and Variant Definitions

Purpose. Define the implementation vocabulary used in the paper and record one exact final recipe per dataset family.

B.1 Where adaptation is applied

Selective adaptation is inserted only in the upper portion of the pretrained CBraMod backbone. Lower layers remain inherited from the pretrained CBraMod backbone. This keeps the pretrained encoder as the primary representation engine while restricting extra adaptation capacity to upper layers where downstream specialization appears most useful.

Backbone recap beyond the main text. CBraMod operates on a channel–patch grid. Its patch embedder mixes temporal and coarse spectral structure before the transformer stack, and each criss-cross transformer block separates channel-wise and patch-wise interactions before recombining them. These inherited design choices motivate why our adaptation mechanism is organized around depth-aware and branch-aware specialization rather than around an unstructured adapter.

B.2 Final reported configuration by dataset

Table 7 lists the exact recipe associated with each headline main-paper result. Exploratory alternatives and context variants are separated into Appendix D so that this table is not read as a tuning sweep. Table 7 gives the compact recipe name associated with each headline result. Full implementation definitions are provided in Table 8, while exploratory alternatives are reserved for Appendix D. The table is therefore intended as a provenance map for main-paper values rather than as a hyperparameter sweep.

Table 7: Final reported configuration by dataset. Each row lists the exact recipe associated with the headline main-paper result for that dataset family; exploratory alternatives are reported separately in Appendix D.

Dataset	Adapted	AttnRes	Router/context	Ctx.	Reg./LR	Sel.
FACED	top 1	pre-attn	block-shared	none	—	val κ
SEED-V	top 1	pre-attn	dual-query	none	—	val κ
ISRUC	top 1	pre-attn	dual-query	none	—	val κ
TUEV	top 1	pre-attn	compact + Δ attn	EEG summary	—	val κ
PhysioNet-MI	top 1	pre-attn	compact-shared	none	bank-reg + comp-LR	val κ

B.3 Variant dictionary

Table 8 defines the compact implementation tags used in the final-recipe and ablation tables. These tags are used only to identify implementation variants; performance interpretation is deferred to Appendices C and D.

Table 8: Variant dictionary. This table maps paper terminology to implementation-level meaning.

Term	Meaning
Dense finetuning	Task head plus full pretrained backbone trainable
AttnRes-only	Backbone with AttnRes-enhanced depth aggregation, no specialist bank
MoE-only	Typed specialists enabled without depth-router features
Full selective	AttnRes, typed specialists, and depth-aware routing enabled
Block-shared	Implementation tag <code>block-shared</code> ; shared routing context across typed branches
Dual-query	Implementation tag <code>dual-query</code> ; separate branch-specific routing queries or context streams
Compact-shared	Implementation tag <code>compact-shared</code> ; compact context shared by router branches
Δ attn	AttnRes-derived attention/residual depth-summary variant (<code>attn_delta4</code> in implementation)
comp-LR	Component-wise learning-rate adjustment used in focused boundary-case runs
bank-reg	Expert-bank regularization used in focused boundary-case runs
EEG summary	Compact sample-level EEG descriptor used as auxiliary router context
PSD summary	Compact spectral descriptor used as auxiliary router context

B.4 Baseline trainability definitions

The baseline definitions are organized by trainable modules and interpretive role. This helps distinguish standard dense finetuning from depth aggregation, typed specialists, and the full selective adaptation stack.

Table 9: Baseline trainability definitions. Each baseline has a specific interpretive role rather than being included only for leaderboard breadth.

Baseline	Trainable modules	Interpretive purpose	Main-paper role
Frozen backbone + head	task head only; full backbone frozen	lower-bound transfer control	no
Dense finetuning	task head plus full backbone	standard downstream baseline	yes
Top- k finetuning	task head plus selected upper layers	simple selective control	no
AttnRes-only	task head plus AttnRes-enhanced backbone path	isolates depth aggregation	yes
MoE-only	task head plus specialists without depth-router features	isolates specialist capacity	yes
Full selective	task head plus AttnRes, specialists, and depth router	full proposed method	yes
Adapter / LoRA baseline	lightweight inserted adaptation modules	generic PEFT contextual control	no

All compared methods within a dataset share the same prediction-head family so that the benchmark isolates adaptation strategy rather than classifier redesign. Dataset-specific head choices and run-level

reporting conventions are documented again in Appendix F together with the result-provenance summary.

C Seed-Complete Results and Stability

Purpose. Separate headline κ -selected values from seed-complete stability summaries so that the main-paper claims can be read with the right level of confidence. This section reports the headline values used in the main paper together with seed-complete and per-seed summaries, while preserving a clear distinction between confirmatory and supportive evidence.

C.1 Reporting convention: headline vs. seed-complete stability

The main paper reports headline values from the κ -selected checkpoint associated with each reported method family. This appendix separates those headline values from seed-complete stability summaries. Mean $\pm$ std values are computed across available seeds for seed-complete families, while FACED/ISRUC payoff rows and focused PhysioNet-MI ablations are labeled as supportive single-run or focused evidence. This separation prevents best-run headline values, multi-seed stability summaries, and exploratory ablations from being interpreted as the same evidentiary object.

C.2 SEED-V reproduced results

Table 10: SEED-V reproduced results. Headline columns report the κ -selected values used in the main paper. Mean columns summarize seed-complete stability across the listed seeds and are not used to replace the headline entries.

Method	Head BA	Head κ	Head wF1	Mean BA	Mean κ	Mean wF1
Dense	0.4024	0.2598	0.4111	0.4019 $\pm$ 0.0005	0.2581 $\pm$ 0.0014	0.4091 $\pm$ 0.0020
AttnRes-only	0.4081	0.2678	0.4164	0.4067 $\pm$ 0.0018	0.2658 $\pm$ 0.0020	0.4145 $\pm$ 0.0017
MoE-only	0.4095	0.2695	0.4174	0.4072 $\pm$ 0.0024	0.2662 $\pm$ 0.0029	0.4153 $\pm$ 0.0022
Full-block	0.4113	0.2710	0.4178	0.4075 $\pm$ 0.0036	0.2664 $\pm$ 0.0050	0.4148 $\pm$ 0.0031
Full-dual	0.4162	0.2805	0.4257	0.4077$\pm$0.0019	0.2668$\pm$0.0026	0.4151$\pm$0.0023

The SEED-V headline ladder supplies the values used in Table 3 and Figure 3. The mean $\pm$ std columns provide the seed-complete stability view. This distinction is important because the headline Full-dual value is the strongest κ -selected run used in the main paper, whereas the mean columns summarize stability across available seed-complete runs.

C.3 TUEV reproduced results

Table 11: TUEV reproduced results. Headline columns report the κ -selected values used in the main paper. Mean columns summarize seed-complete stability across the listed seeds and are not used to replace the headline entries.

Method	Head BA	Head κ	Head wF1	Mean BA	Mean κ	Mean wF1
Dense	0.6213	0.6540	0.8188	0.6040 $\pm$ 0.0231	0.6110 $\pm$ 0.0578	0.7982 $\pm$ 0.0272
AttnRes-only	0.6115	0.6235	0.8108	0.5949 $\pm$ 0.0234	0.6168 $\pm$ 0.0095	0.8062 $\pm$ 0.0047
Full selective	0.6798	0.6480	0.8189	0.6196$\pm$0.0391	0.5979$\pm$0.0467	0.7933$\pm$0.0223

TUEV shows the same distinction between headline and stability views. The full selective recipe gives the strongest headline BA under the κ -selected protocol, but the seed-complete mean view is more mixed. We therefore treat TUEV as matched control evidence for a strong headline BA improvement, not as evidence for uniformly better performance across every metric and seed.

C.4 Headline run provenance used in main tables

The archived logs contain the headline test metrics used in Tables 2–3. All headline runs follow the validation- κ selection protocol defined in Table 1 and Appendix A. Full log paths and validation-summary exports are provided in the anonymized release artifact rather than printed in the table to avoid unreadable table wrapping.

Table 12: Headline run provenance for main-paper values. All rows use validation κ selection under the paper protocol. Log identifiers are shortened for readability; full paths are provided in the anonymized release artifact.

Dataset	Variant	Run ID	Ckpt	Test BA	Test κ	Test wF1
SEED-V	Full-dual	1769784	18	0.4162	0.2805	0.4257
TUEV	Full selective	1539694	1	0.6798	0.6480	0.8189
FACED	Full block-shared	1464851	32	0.6055	0.5528	0.6072
ISRUC	Full dual-query	1521786	18	0.8068	0.7802	0.8306

C.5 Per-seed values

The per-seed rows below support the mean $\pm$ std summaries above. They are included for auditability rather than as additional leaderboard comparisons.

Table 13: SEED-V per-seed values supporting the seed-complete summaries in Table 10. These rows are included for auditability and are not additional leaderboard entries.

Method	Seed	BA	κ	wF1
Dense	42	0.40244	0.25975	0.41106
Dense	3407	0.40116	0.25806	0.40987
Dense	2024	0.40199	0.25641	0.40634
AttnRes	42	0.40809	0.26776	0.41641
AttnRes	3407	0.40418	0.26308	0.41220
AttnRes	2024	0.40792	0.26643	0.41481
MoE-only	42	0.40950	0.26953	0.41743
MoE-only	3407	0.40389	0.26237	0.41229
MoE-only	2024	0.40821	0.26658	0.41604
Full-block	42	0.41127	0.27103	0.41778
Full-block	3407	0.40867	0.26864	0.41604
Full-block	2024	0.40263	0.25952	0.41061
Full-dual	42	0.41023	0.27036	0.41792
Full-dual	3407	0.40581	0.26438	0.41220
Full-dual	2024	0.40699	0.26579	0.41505

Table 14: TUEV per-seed values supporting the seed-complete summaries in Table 11. These rows are included for auditability and are not additional leaderboard entries.

Method	Seed	BA	κ	wF1
Dense	42	0.62130	0.65401	0.81884
Dense	3407	0.57784	0.54528	0.76739
Dense	2024	0.61292	0.63384	0.80840
AttnRes	42	0.60500	0.62100	0.81076
AttnRes	3407	0.61148	0.60604	0.80141
AttnRes	2024	0.56811	0.62351	0.80632
Full	42	0.65984	0.64527	0.81467
Full	3407	0.58167	0.55184	0.77009
Full	2024	0.61725	0.59649	0.79504

C.6 Supportive payoff results

FACED and ISRUC are reported as supportive payoff evidence because the strongest current runs are not a fully seed-complete matched matrix. These rows correspond to the practical payoff block in Table 2 and are not used as confirmatory component evidence.

Table 15: Supportive payoff rows for FACED and ISRUC. These single-run rows correspond to the practical payoff evidence summarized in Table 2 and are not treated as confirmatory component evidence.

Dataset	Variant	BA	κ	wF1
FACED	Full block-shared typed	0.6055	0.5528	0.6072
ISRUC	Full dual-query typed	0.8068	0.7802	0.8306

D Extended Ablations and Boundary-Case Evidence

Purpose. Collect the additional variants that sharpen the dataset-dependent interpretation without repeating the confirmatory control evidence.

D.1 Context and depth-summary variants

Table 16 collects support variants that are not used as the main causal evidence but help explain why the effect of routing context is dataset-dependent. These rows are single-run supportive evidence unless explicitly marked as seed-complete and should not be read as a separate leaderboard.

Table 16: Context and depth-summary variants from the current logs. FACED and ISRUC rows are single-run supportive variants. SEED-V rows report seed-complete means without standard deviations for compact comparison; the full mean $\pm$ std stability view is reported in Appendix C.

Dataset	Variant	BA	κ	wF1
FACED	compact-shared	0.5808	0.5246	0.5826
FACED	dual-query	0.5841	0.5288	0.5857
FACED	block-shared	0.6055	0.5528	0.6072
ISRUC	compact+ Δ attn	0.7959	0.7724	0.8260
ISRUC	dual-query	0.8068	0.7802	0.8306
SEED-V	block-shared	0.4075	0.2664	0.4148
SEED-V	dual-query	0.4077	0.2668	0.4151

Table 16 summarizes context and depth-summary variants without treating them as a second benchmark table. FACED and ISRUC show that typed or dual-query context can be useful in payoff regimes, whereas SEED-V shows only small differences between block-shared and dual-query routing when summarized across the seed-complete runs. Standard deviations for SEED-V are omitted here for compactness and reported in Appendix C. Overall, these variants support the interpretation that routing/context design is dataset-dependent rather than uniformly dominant.

D.2 PhysioNet-MI boundary-case ablation

Table 17: PhysioNet-MI boundary-case ablation. All rows are single-run supportive evidence and should be interpreted as focused boundary analysis rather than as confirmatory benchmark results.

Group	Variant	BA	κ	wF1
Reg./LR	none	0.6008	0.4676	0.6004
Reg./LR	bank-reg	0.6014	0.4684	0.6020
Reg./LR	comp-LR	0.5925	0.4565	0.5911
Reg./LR	bank-reg + comp-LR	0.6129	0.4838	0.6122
Compact ctx.	none	0.6002	0.4668	0.5982
Compact ctx.	EEG summary	0.5847	0.4461	0.5840
Compact ctx.	PSD summary	0.5952	0.4602	0.5946

PhysioNet-MI constrains the generality claim. Under the current κ -selection protocol, the regularization/LR block shows that bank regularization plus mild component learning-rate adjustment gives the strongest focused recipe, while the compact-feature block shows that EEG-summary and PSD-summary context do not improve over the no-context reference. Because these are focused

single-run ablations, they support the boundary-case interpretation but are not confirmatory multi-seed benchmark evidence.

E Routing Diagnostics and Mechanism Support

Purpose. Provide mechanism-supporting evidence about expert utilization without turning the diagnostics into a second performance benchmark.

E.1 Diagnostic definitions

We use routing diagnostics only to ask whether the specialist pathway is nontrivially used. Effective experts, N_{eff} , summarizes how many experts receive meaningful routing mass; normalized entropy, $H/\log E$, measures how diffuse routing is relative to the number of experts; and top-expert share measures concentration on the most-used expert. These diagnostics are descriptive mechanism evidence and are not used as standalone proof of downstream performance gains:

$$N_{\text{eff}} = \exp(H(p)), \quad H(p) = -\sum_{j=1}^E p_j \log p_j, \quad H_{\text{norm}} = H(p)/\log E.$$

E.2 Expert-utilization and entropy table

Table 18: Representative routing diagnostics extracted from current log exports. Values summarize one diagnostic snapshot per reproduced family and support the claim that specialist paths are nontrivially used. They are descriptive mechanism evidence rather than standalone proof of downstream performance gains.

Dataset	Variant	Branch	N_{eff}	$H/\log E$	Top expert share
SEED-V	MoE-only	spatial	2.746	0.729	0.656
SEED-V	MoE-only	spectral-temporal	2.776	0.736	0.484
SEED-V	Full-block	spatial	2.102	0.536	0.719
SEED-V	Full-block	spectral-temporal	2.688	0.713	0.656
SEED-V	Full-dual	spatial	3.528	0.909	0.344
SEED-V	Full-dual	spectral-temporal	2.020	0.507	0.750
TUEV	Full	spatial	1.938	0.477	0.625
TUEV	Full	spectral-temporal	1.755	0.406	0.750

E.3 Routing diagnostic plots

E.4 Interpretation and limitations

The available SEED-V and TUEV logs show nontrivial, branch-dependent expert utilization. SEED-V generally uses more experts than TUEV, and the spatial and spectral-temporal branches can behave differently within the same run family. The strongest SEED-V full variant shows high spatial diversity but sharper spectral-temporal routing, consistent with complementary rather than redundant specialization.

These diagnostics are useful as mechanism support but not as causal proof. They show that the selective path is not trivially unused and that branch behavior differs across datasets, but the performance claims remain grounded in the matched downstream comparisons reported in the main paper and Appendix C.

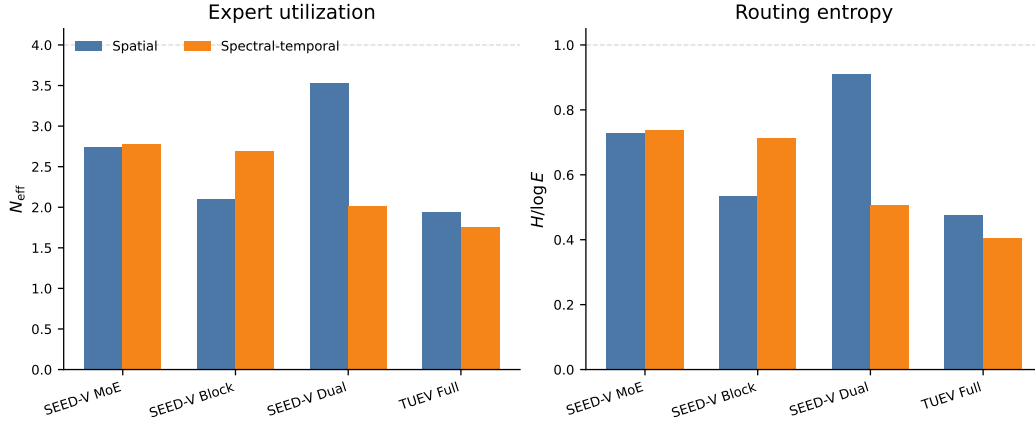


Figure 5: Representative routing diagnostics across reproduced families. The left panel shows expert utilization via N_{eff} , and the right panel shows normalized routing entropy via $H/\log E$. Variant ordering matches the main-paper Figure 3.

F Reproducibility Details

Purpose. Record the training settings, seed conventions, compute envelope, and result-provenance summary needed to audit the reported evidence without tying the paper text to repository artifacts.

F.1 Training settings

The following table summarizes optimization and adaptation settings for the reported families. Settings are held fixed within each dataset family, making the within-dataset comparisons auditable.

Table 19: Optimization and adaptation settings for the reported families. Settings are held fixed within each family, which makes the within-dataset comparisons auditable.

Dataset / family	Epochs	Batch	LR	Min LR	WD	EMA	Dispatch	Selection / role
FACED selective family	40	32	2e-4	–	2e-2	on	soft	payoff variants; val κ
SEED-V family	25	64	3e-5	5e-6	3e-2	on	soft	confirmatory control family; val κ
ISRUC selective family	30	16	3e-5	–	2e-2	on	soft	payoff variants; val κ
TUEV dense family	20	8	5e-5	5e-7	2e-2	off	–	dense control; val κ
TUEV full family	20	8	5e-5	5e-7	2e-2	on	soft	confirmatory control family; val κ
PhysioNet-MI focused family	30	64	3e-5	1e-6	2e-2	off	soft	boundary-case study; val κ

F.2 Seeds and checkpoint selection

For seed-complete families, runs use seeds 42, 3407, and 2024. Across all datasets, checkpoint selection uses validation Cohen’s κ as the prespecified selection rule in the current protocol. Headline values correspond to the κ -selected checkpoint associated with each reported method family, while mean $\pm$ std values are computed across the available seeds for seed-complete families. The same seed set is used for the seed-complete control families; single-run support families are labeled explicitly in the relevant result tables.

F.3 Compute and execution envelope

Table 20 reports log-derived compute for retained completed runs. Because some archived scheduler logs do not consistently preserve physical GPU model and capacity, we report observed peak CUDA allocation separately from wall-clock time and GPU-hour estimates. Retained runs account for approximately 261.8 GPU-hours, while preliminary, failed, or incomplete runs account for an

Table 20: Compute envelope estimated from archived retained-run logs. Times are median wall-clock durations per completed run. Peak CUDA allocation is reported when available from logs; it is not a substitute for physical GPU capacity.

Family	Runs	Peak alloc. (GB)	Time/run	GPU-h	Notes
FACED	54	14.44	24.0 min	23.8	payoff/support variants
SEED-V	63	8.83	30.9 min	37.1	dense, AttnRes, MoE, full families
ISRUC	19	56.95	120.3 min	34.2	payoff/support variants
TUEV	64	1.06	138.2 min	150.9	dense and full selective families
PhysioNet-MI	75	17.61	7.8 min	15.8	focused boundary-case variants
Total retained	275	–	–	261.8	completed retained runs
Prelim./failed	–	–	–	24.1	development/debugging envelope

510 additional approximate 24.1 GPU-hours. Preliminary runs were used for development, debugging, or
511 ablation screening and are not treated as primary evidence unless explicitly reported.

512 **F.4 Result provenance summary**

513 The reported tables and figures are backed by structured experimental exports, but the paper does not
514 need to enumerate repository filenames. For auditability, the retained records should preserve three
515 evidence groups: seed-complete benchmark exports that support the headline reproduced values,
516 mean $\pm$ std summaries, and per-seed tables in Appendix C; boundary-case and support-run exports
517 that support the FACED/ISRUC payoff rows and the focused PhysioNet-MI ablations in Appendix D;
518 and routing-diagnostic exports that support the expert-usage, entropy, and branch-level summaries in
519 Appendix E.

520 **F.5 Code release and data access**

521 **Code release and data access.** The submission includes an anonymized code artifact for review,
522 provided as a repository snapshot or supplemental archive, with training and evaluation scripts,
523 configuration files, preprocessing instructions, and commands for reproducing the reported tables.
524 This work uses five existing EEG benchmarks—FACED, SEED-V, ISRUC, TUEV, and PhysioNet-
525 MI—and the pretrained CBraMod backbone/codebase. We cite the original dataset and model sources
526 in Section 4.1 and provide source/access links in the released documentation. Raw benchmark EEG
527 data and pretrained checkpoints are not redistributed when the original license, data-use agreement,
528 or access policy does not permit redistribution. Instead, the artifact documents how to obtain each
529 dataset and model asset from its original source and how to run preprocessing and evaluation locally.

530 **Responsible use and broader impact.** This work may help make pretrained EEG backbones more
531 adaptable across heterogeneous downstream settings, potentially reducing the need to train separate
532 task-specific models from scratch and improving reproducibility in EEG transfer studies. At the same
533 time, EEG data can contain sensitive neural and health-related information, and improved adaptation
534 methods should not be interpreted as clinical, diagnostic, affective-state, or BCI deployment readiness
535 without external validation, appropriate consent, and dataset-specific governance. The released
536 materials therefore provide code and data-access instructions rather than redistributing restricted raw
537 EEG data, and all benchmark use should follow the original dataset licenses, consent constraints, and
538 access policies. We also caution that performance differences observed on public benchmarks may
539 not generalize across populations, recording systems, clinical groups, or deployment contexts.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Introduction frame the contribution as a downstream adaptation framework for a pretrained EEG backbone, not a new pretraining method or foundation model (Section 1). Section 5 then qualifies claims using contextual payoff evidence (FACED/ISRUC), matched reproduced control evidence (SEED-V/TUEV), and supportive boundary evidence (PhysioNet-MI).

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6.1 discusses backbone specificity, heterogeneous evidence strength across datasets, and the more tentative status of routing-mechanism conclusions. Additional scope qualifications are provided in Section 5.5 and Appendices C–E.

Guidelines:

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